

Hydrogeochemical evidence for biotic over volcanic control in Lake Maninjau, Indonesia

Taofik Jasalesmana¹⁾ , Mutiara Rachmat Putri^{*2)} , Mirzam Abdurrachman³⁾ ,
Cynthia Henny⁴⁾ , Sulung Nomosatryo⁴⁾ 

¹⁾ Bandung Institute of Technology, Faculty of Earth Sciences and Technology, Earth Science Doctor Program,
Jl. Ganesa No. 10 Coblong, 40132 Bandung, Jawa Barat, Indonesia

²⁾ Bandung Institute of Technology, Faculty of Earth Sciences and Technology,
Research Group of Environmental and Applied Oceanography, Jl. Ganesa No. 10 Coblong, 40132 Bandung, Indonesia

³⁾ Bandung Institute of Technology, Faculty of Earth Sciences and Technology, Geological Engineering Study Program,
Jl. Ganesa No. 10 Coblong, 40132 Bandung, Indonesia

⁴⁾ National Research and Innovation Agency (BRIN), Research Center for Limnology and Water Resources,
Gedung B.J. Habibie, Jl. M.H. Thamrin No. 8, Jakarta Pusat 10340, Indonesia

* Corresponding author

RECEIVED 28.11.2025

ACCEPTED 23.03.2026

AVAILABLE ONLINE 17.09.2026

Abstract: Lake Maninjau, a volcanic caldera lake in Indonesia, frequently experiences sulphide upwelling events that trigger periodic fish kills. Although these events are commonly attributed to eutrophication and organic matter enrichment, the potential contribution of residual volcanic activity remains unclear. This study re-evaluates the hydrogeochemical characteristics of Lake Maninjau to distinguish volcanic signatures from biotic sulphide generation. Field observations were conducted at four sites representing contrasting environments: a dome-proximal area with presumed volcanic influence (DM3), a eutrophic bay affected by intensive aquaculture (DM1), and two additional stations (DM2 and DM4) representing transitional and deep-water reference sites. Temperature, pH, total dissolved solids (TDS), and major anions (HCO_3^- , Cl^- , SO_4^{2-} , HS^-) were measured in the water column and porewater, while sedimentary sulphur and carbon contents were analysed to identify sulphide sources. The results showed no thermal or ionic anomalies near the dome-proximal area. Across all sites, the waters exhibited neutral to alkaline pH (6.5–8.9), low TDS ($<0.11 \text{ g}\cdot\text{dm}^{-3}$), and bicarbonate dominance, indicating a carbonate-buffered system rather than magmatic or hydrothermal influence. Hydrogen sulphide accumulated in the lower water column, reaching a maximum concentration of $0.401 \text{ mg}\cdot\text{dm}^{-3}$. A strong positive correlation between sediment sulphur and carbon (Pearson's $r = 0.73$, $p = 0.01$), together with positive intercepts in C–S plots, indicates that sulphide production is primarily driven by microbial sulphate reduction associated with organic matter degradation. These findings demonstrate that sulphide accumulation in Lake Maninjau is predominantly biotic, highlighting the role of internal biogeochemical processes rather than residual volcanic activity.

Keywords: biogenic sulphur, biogeochemistry, hydrogeochemistry, Lake Maninjau volcanic dome

INTRODUCTION

Volcanic crater and caldera lakes are unique hydrological systems shaped by the interplay of geological and biogeochemical processes. In active volcanic settings, the continuous release of

magmatic gases and hydrothermal fluids often exerts a dominant influence on lake water chemistry, resulting in acidic pH, elevated ionic strength (Varekamp, Pasternack and Rowe Jr., 2000), and high concentrations of dissolved species such as Cl^- , SO_4^{2-} and H_2S (Rouwet *et al.*, 2014). These features make volcanic lakes

important natural analogues for understanding magmatic degassing, fluid–rock interactions, and geochemical hazards associated with limnic eruptions (Kusakabe, 2015).

However, in caldera lakes that have entered a dormant or post-volcanic phase, the predominant geochemical regime often shifts from magmatic control to biologically mediated processes. In these systems, internal biogeochemical cycling, particularly microbial sulphate reduction, can become the primary source of sulphide accumulation, independent of volcanic inputs (Duarte and Prairie, 2005; Henny and Nomosatryo, 2016). These processes are typically enhanced by anthropogenic nutrient enrichment, increased organic matter deposition, and prolonged water-column stratification, all of which favour suboxic or anoxic conditions in the hypolimnion (Kankaala *et al.*, 2013; Nomosatryo *et al.*, 2021). Recognising this transition is essential for understanding how volcanic lakes evolve under natural and human-induced pressures.

Lake Maninjau, located in West Sumatra, Indonesia, is a caldera lake formed approximately 52,000 years ago by a catastrophic volcanic eruption (Alloway *et al.*, 2004). The lake retains several volcanic features, including a central dome that has been proposed as a conduit for residual geothermal input (Wils *et al.*, 2021). However, in recent decades, eutrophication has increased due to aquaculture activities, leading to periodic fish-kill events linked to hypoxia and sulphide upwelling (Syandri *et al.*, 2014; Fukushima *et al.*, 2017).

While previous studies have described oxygen depletion, nutrient accumulation, and biotically driven sulphide dynamics in Lake Maninjau (Henny, 2009; Henny and Nomosatryo, 2012), there has been no comprehensive hydrogeochemical assessment to determine whether the current sulphur and carbon processes

remain influenced by residual volcanic activity or are now primarily controlled by biological and anthropogenic inputs. Earlier studies focused on the ecological and biogeochemical consequences of sulphide production but did not explicitly evaluate the potential volcanic contributions to the present-day geochemical system of the lake. This distinction is critical for evaluating lake management strategies and understanding how tropical volcanic lakes evolve from magmatic to biologically dominated systems under increasing anthropogenic pressure. Therefore, this study aimed to provide hydrogeochemical evidence for the dominant biotic control of the geochemical system of Lake Maninjau. By comparing the chemical and sediment characteristics of two contrasting environments, this study evaluated the relative contributions of volcanic and biotic processes: (1) a dome-proximal area presumed to reflect possible volcanic influence and (2) an organically enriched eutrophic area affected by aquaculture inputs. These findings provide new insights into the transformation of tropical volcanic lakes from magmatic to biological dominance and highlight the growing role of human-induced organic loading in shaping lake biogeochemistry.

MATERIALS AND METHODS

STUDY SITE

Lake Maninjau, located in West Sumatra Province, Indonesia, is a volcanic caldera lake with an average depth of 105 m, a maximum depth of 165 m, a surface area of 99.5 km², and a volume of 10.1 km³ (Santoso *et al.*, 2018) – Figure 1. The lake is bordered by lahar deposits from the Maninjau caldera, particu-

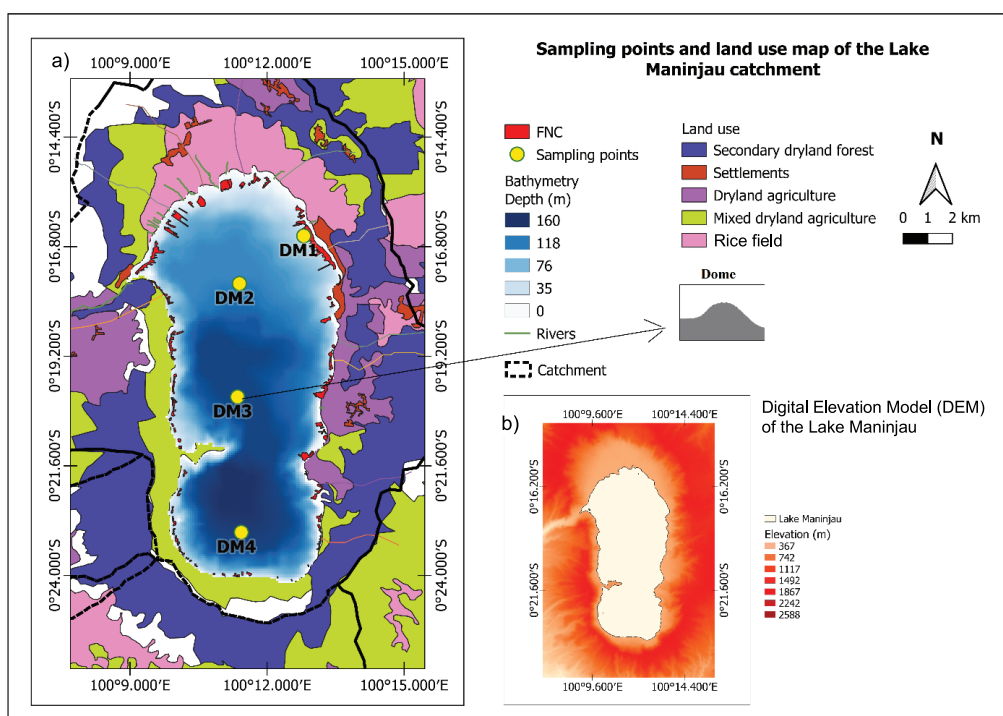


Fig. 1. Map of the Lake Maninjau catchment, West Sumatra, Indonesia: a) sampling locations and land use, b) the digital elevation model (DEM) of the Lake Maninjau area derived from DEMNAS; DM1–DM4 = stations, FNC = floating net cages; source: own elaboration based on land use data obtained from the Indonesian Geospatial portal (<https://www.indonesia-geospasial.com/>) from the Ina-Geoportal (<https://tanahair.indonesia.go.id/portal-web/unduh/demnas>); distribution of floating net cages digitised from Google Earth imagery

larly in the north-western, western, and south-eastern regions (Pribadi, Mulyadi and Pratomo, 2007). Steep cliffs, particularly along the southern wall, were formed by faulting and volcanic eruptions. Most rivers flowing into the lake are intermittent, with the Batang Antokan River serving as the outflow of the lake and draining through the north-west wall into the Indian Ocean (Pribadi, Mulyadi and Pratomo, 2007; Setiawan *et al.*, 2019). However, spatial differences in catchment topography influence the water quality of Lake Maninjau. The northern to eastern catchment, characterised by gentler slopes and extensive rice fields and settlements, is the main source of runoff and dissolved organic nitrogen (DON) to the lake under various hydrological conditions, whereas the steeper, shrub- and forest-covered western catchment contributes lower nutrient loads (Rustini, Harsono and Ridwansyah, 2019).

The limnological and biogeochemical conditions of Lake Maninjau have been profoundly influenced by intensive aquaculture, particularly the extensive use of floating net-cage systems (Ind.: Keramba Jaring Apung – KJA) (Rustini, Harsono and Ridwansyah, 2019). Based on official records from the Department of Food Security and Fisheries (Ind.: Dinas Ketahanan Pangan dan Perikanan – DKPP) of Agam Regency, Indonesia, 23,359 floating net cages were in active operation in the lake in 2024 (DKPP, 2024). This number exceeds the estimated aquaculture carrying capacity in recent years (Makmur, Muth-mainnah and Subagdja, 2020). These systems represent a significant source of anthropogenic nutrient loading, as substantial quantities of nitrogen and phosphorus derived from fish feed residues, fish excreta, and sewage associated with net cage operations are released directly into the water column (Syandri *et al.*, 2014). Consequently, excessive nutrient enrichment has been widely linked to eutrophication, hypoxia, and the accumulation of organic matter in both the water column and sediments of Lake Maninjau (Santoso and Triwisesa, 2020; Sulastri, Aisyah and Dina, 2022; Jasalesmana, Putri and Abdurrachman, 2023).

WATER QUALITY MEASUREMENT AND WATER COLLECTION

Field surveys were conducted from 26 November to 1 December 2022, and from 24 to 29 August 2023. The first survey coincided with a rainfall-dominated period, whereas the second survey was conducted under dry conditions (Putri *et al.*, 2024). Measure-

ments of physicochemical parameters (temperature, pH, and total dissolved solids – TDS) and water sampling were carried out at four stations (DM1, DM2, DM3, and DM4) – Figure 1. Detailed descriptions of each station are presented in Table 1. The temperature profile in November 2022 was measured continuously (at 20 cm intervals) using a CTD Rinko Profiler (JFE Advantech Co., Ltd., Tokyo, Japan).

In August 2023, temperature profiles were measured discretely using a multiparameter water quality probe (HORIBA U-52; HORIBA Advanced Techno Co., Ltd., Kyoto, Japan) owing to the unavailability of the CTD profiler. Although the HORIBA U-52 provides a lower vertical resolution than the CTD profiler, temperature measurements were conducted carefully at predefined depth intervals, which were sufficient to characterise the overall thermal structure of the water column. Other physicochemical parameters (TDS and pH) in both November 2022 and August 2023 were measured using a Series Multi-Parameter Water Quality HORIBA U52 (HORIBA Advanced Techno, Co., Ltd, Kyoto, Japan). In each survey, water samples were collected using a Niskin water sampler for 5 dm³ sample (General Oceanics, Miami, FL, United States) from five depth strata at intervals adjusted according to the water column temperature and total water depth. Specifically, the sampling depths at DM4 were 0, 2, 10, 60, and 158 m; at DM3 were 0, 2, 10, 60, and 144 m; at DM2 were 0, 2, 10, 60, and 110 m; and at DM1, sampling was conducted at 0, 2, 10, and 30 m owing to the shallow water depth. Water samples were collected by drawing water directly from the water sampler using a 50 cm³ syringe. The water was filtered using 0.45-µm cellulose acetate membrane syringe filters and then transferred (10 cm³) into 15 cm³ test tubes for anion analysis. Approximately 10 cm³ of the water column sample was preserved for sulphide analysis immediately after collection by adding 5% zinc acetate and sodium hydroxide (NaOH). Zinc acetate was added to quantitatively bind the dissolved sulphide as zinc sulphide (ZnS), thereby preventing oxidation by atmospheric oxygen prior to analysis.

SEDIMENT CORE IN DM3 AND DM1 STATIONS

Sediment samples were collected at stations DM3 and DM1 using a gravity corer (90 cm in length and 7 cm in internal diameter). These stations were selected to represent contrasting environmental settings, with DM3 being predominantly influenced by

Table 1. Stations descriptions at Lake Maninjau

Station	Description
DM1	a shallow (35 m), eutrophic bay on the north-eastern lakeshore, is heavily impacted by floating net cages; the 3,354 fish cages release a continuous load of organic waste from feed and faeces (Junaidi <i>et al.</i> , 2022), making DM1 an area dominated by organic and biotic processes
DM2	in the middle-western part of the lake (~110 m depth), an intermediate zone between shallow eutrophic and deep volcanic environments; underlying fault lines (Wils <i>et al.</i> , 2021), suggest possible minor geothermal inputs, influencing its unique physical and chemical conditions
DM3	near the central part of the lake with a depth of approximately 136 m, located above a dome-like structure, which has been suspected as a potential indicator of residual or low-level volcanic activity (Wils <i>et al.</i> , 2021); therefore, DM3 represents the volcanic-influence test site, used to verify whether any remaining geothermal contribution affects sulphide dynamics
DM4	the deepest part of the lake (~158 m), a reference site for its natural deep-water environment; permanent stratification and anoxic bottom waters, providing baseline geochemical conditions unaffected by direct anthropogenic or volcanic input

Source: own elaboration.

volcanic processes and DM1 being characterised by stronger anthropogenic influences (Tab. 1).

At station DM3, the sediment core was sectioned at 2 cm intervals from the surface to a depth of 16 cm to obtain a high-resolution vertical geochemical profile. In contrast, sediment sectioning at station DM1 was limited to selected depth intervals: the surface layer (0–2 cm) and two deeper sections at depths of 20.0–22.5 cm and 25.0–27.5 cm. This sectioning strategy was designed to capture key sedimentary domains relevant to carbon and sulphur cycling at each site, with the surface sediment representing recent depositional conditions, whereas the deeper layers reflect earlier depositional phases and more stabilised sedimentary environments. All sediment samples were sealed in nylon plastic bags (12 × 20 cm), enclosed in gas-tight aluminium foil bags, and stored at –18°C prior to laboratory analyses to preserve sample integrity.

ANION ANALYSIS

Major inorganic anions (Cl^- and SO_4^{2-}) were analysed using suppressed ion chromatography on a Dionex ICS 6000, Thermo Fisher Scientific, using a Dionex IonPac AS14 column set. The eluent was 20 mM KOH (cartridge). The flow rate was set to $1.5 \text{ cm}^3 \cdot \text{min}^{-1}$ and the column temperature was set to 30°C. A seven-anion calibration standard from Dionex, Thermo Fisher, was used to analyse the anions. The sulphide concentration was determined using the methylene blue method (HACH method 8131; method detection limit – MDL = $5 \mu\text{g} \cdot \text{dm}^{-3}$). Alkalinity was determined using a titration method with 0.1 N H_2SO_4 as the titrant. The titration endpoint was set at pH 4.5, which was measured using a calibrated pH meter (Pearson, 1981). The alkalinity values obtained were used to calculate the bicarbonate (HCO_3^-) concentration.

CARBON, SULPHUR, AND TRACE ELEMENT ANALYSIS IN SEDIMENTS

The carbon and sulphur contents were determined using sediment samples from stations DM3 and DM1. Approximately 3 g of sediment from each selected depth interval was freeze-dried for 48 h and analysed using a FlashSmart elemental analyser (Model 159951) to determine total carbon and sulphur concentrations.

For trace element analysis, approximately 1 g of oven-dried sediment from each depth interval at DM3 and DM1 was finely ground and pressed into pellets. The elemental composition was measured using a Bruker S2 Puma XRF spectrometer equipped with a Pd anode X-ray tube (maximum 50 kV, 2 mA) and automatic primary filters (10 positions) to optimise excitation conditions. Measurements were conducted in air using a 12 mm collimator mask, with sample positioning and documentation supported by an integrated HD camera. Elemental concentrations, expressed as either ppm to weight percent, were calculated using the manufacturer's proprietary software.

DATA ANALYSIS

Statistical analyses were conducted to evaluate spatial patterns and relationships among the physicochemical and geochemical parameters. One-way analysis of variance (ANOVA) was used to

assess spatial differences among the sampling stations. To account for vertical heterogeneity in the water column, separate ANOVA was performed for samples above and below the thermocline at each station. The thermocline depth was determined from temperature profiles, which were used to distinguish the upper mixed layer from the deeper thermally stratified layer. All ANOVA tests were conducted in RStudio (R version 4.5.1) using the `aov()` function, and when significant differences were detected ($p < 0.05$), post-hoc tests were applied to identify specific station pairs exhibiting significant contrasts.

Interannual differences in total dissolved solids (TDS) were evaluated using independent *t* tests. To investigate the influence of biotic and abiotic processes on sulphur dynamics, a linear regression analysis was performed to examine the relationship between sedimentary carbon and sulphur content.

Principal component analysis (PCA) was employed to explore multivariate relationships among standardised sediment geochemical parameters. The PCA was conducted using the FactoMineR package in R, with the variables scaled to unit variance to ensure equal weighting. The resulting ordination patterns were visualised using biplots generated with the factoextra package, which facilitated the identification of dominant geochemical gradients and station-specific clustering.

The carbon dioxide (CO_2) and bicarbonate ion (HCO_3^-) concentrations were calculated using a carbonate system calculator, CO2Sys.xls, applicable to both seawater and freshwater systems (<https://github.com/jamesorr/CO2SYS-Excel>, last accessed: 14 October 2025). The input parameters included temperature, alkalinity, and pH. To visualise the relative concentrations of sulphate (SO_4^{2-}), chloride (Cl^-), and bicarbonate (HCO_3^-), a ternary diagram was utilised that was performed by the Liquid_Analysis_v1_Powell-2010-StanfordGW.xls spreadsheet (Powell and Cumming, 2010). The concentrations of Cl^- , SO_4^{2-} , and HCO_3^- were entered into the spreadsheet, with all concentrations expressed in $\text{mg} \cdot \text{kg}^{-1}$ (ppm). The Cl^- – SO_4^{2-} – HCO_3^- ternary diagram was automatically generated in the TCSH worksheet, where T denotes ternary and C, S, and H represent chloride, sulphate, and bicarbonate, respectively.

RESULTS

VERTICAL PROFILES OF PHYSICOCHEMICAL PARAMETERS IN THE WATER COLUMN

Vertical profiles of temperature, total dissolved solids (TDS), and pH were measured at four stations (DM1, DM2, DM3, and DM4) during two sampling campaigns in November 2022 and August 2023. The results revealed distinct variations in the vertical distribution of each parameter among the stations, which were influenced by both water column depth and sampling time. An exception was observed for the temperature in November 2022 (Fig. 2), which showed relatively uniform values across the stations. Clear differences between the November 2022 (rainy) and August 2023 (dry) surveys were observed, with temperature and pH generally decreasing with depth, while TDS slightly increased in the deeper layers.

Water column temperatures in November 2022 and August 2023 ranged from 26.79 to 28.93°C and 26.70–29.30°C, respectively, and generally decreased with depth (Fig. 2a). In November

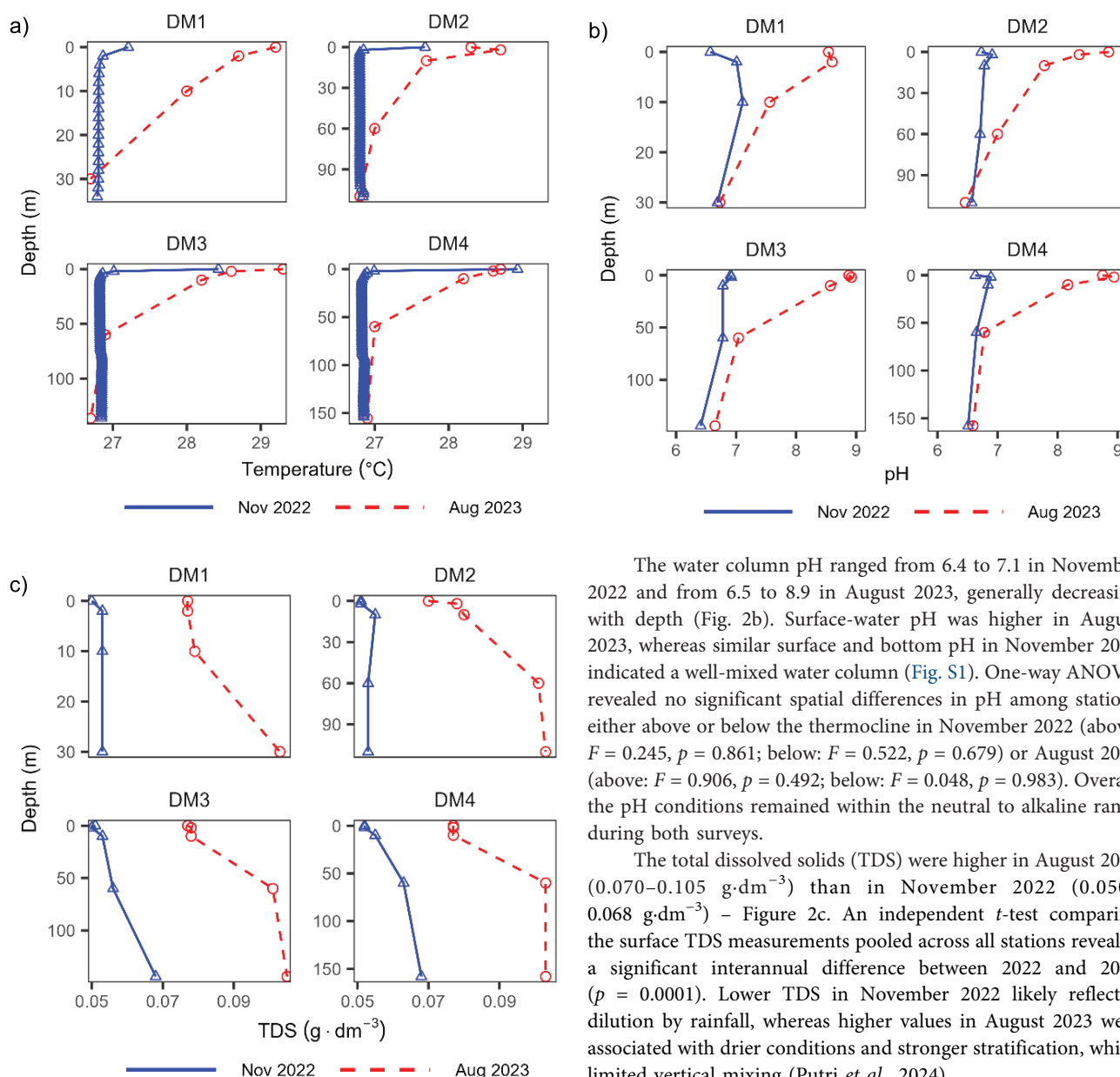
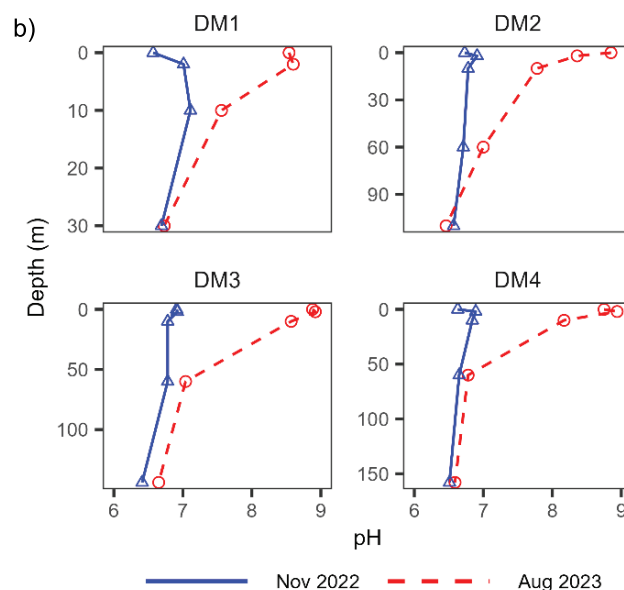


Fig. 2. Water column temperature profile in DM1, DM2, DM3, and DM4 stations: a) temperature, b) pH, c) total dissolved solids (TDS); source: own study

2022, the temperature profile was weakly stratified, with a small surface–bottom temperature difference (ΔT_{sb}) of 1.2°C and a shallow thermocline at approximately 2 m depth, indicating enhanced vertical mixing. In contrast, August 2023 exhibited stronger thermal stratification, characterised by a larger ΔT_{sb} of 2.1°C and a clearly defined thermocline at a depth of approximately 10 m.

Spatial variation in water temperature was observed both above and below the thermocline during the November 2022 survey. In November 2022, one-way ANOVA revealed significant temperature differences among stations both above ($F = 9.346$, $p = 0.000$) and below ($F = 1657$, $p = 0.000$) the thermocline. Post-hoc analysis indicated that, in general, temperatures differed among stations in both layers (Tab. S1). In contrast, in August 2023, no significant spatial differences were detected either above ($F = 0.513$, $p = 0.688$) or below the thermocline ($F = 0.917$, $p = 0.559$).



The water column pH ranged from 6.4 to 7.1 in November 2022 and from 6.5 to 8.9 in August 2023, generally decreasing with depth (Fig. 2b). Surface-water pH was higher in August 2023, whereas similar surface and bottom pH in November 2022 indicated a well-mixed water column (Fig. S1). One-way ANOVA revealed no significant spatial differences in pH among stations either above or below the thermocline in November 2022 (above: $F = 0.245$, $p = 0.861$; below: $F = 0.522$, $p = 0.679$) or August 2023 (above: $F = 0.906$, $p = 0.492$; below: $F = 0.048$, $p = 0.983$). Overall, the pH conditions remained within the neutral to alkaline range during both surveys.

The total dissolved solids (TDS) were higher in August 2023 (0.070–0.105 g·dm⁻³) than in November 2022 (0.050–0.068 g·dm⁻³) – Figure 2c. An independent *t*-test comparing the surface TDS measurements pooled across all stations revealed a significant interannual difference between 2022 and 2023 ($p = 0.0001$). Lower TDS in November 2022 likely reflected dilution by rainfall, whereas higher values in August 2023 were associated with drier conditions and stronger stratification, which limited vertical mixing (Putri *et al.*, 2024).

Relative to the thermocline, no significant spatial differences in TDS were observed among the stations in November 2022 (above: $F = 0.667$, $p = 0.615$; below: $F = 2.184$, $p = 0.168$) or August 2023 (above: $F = 0.626$, $p = 0.624$; below: $F = 0.513$, $p = 0.688$). Vertically, the TDS was nearly uniform at stations DM1 and DM2 in November 2022, indicating effective mixing in shallow areas, whereas a slight increase with depth is found in stations DM3 and DM4. A similar downward increase occurred in August 2023, with overall limited spatial but pronounced temporal variability.

SPATIAL AND TEMPORAL DISTRIBUTION OF MAJOR ANIONS (Cl⁻, SO₄²⁻, HCO₃⁻ AND HS⁻)

The vertical anion profiles at each station during the November 2022 and August 2023 surveys are presented in Figure 3. Across both surveys, sulphate (SO₄²⁻) concentrations generally decreased with depth, whereas chloride (Cl⁻) showed the opposite trend, increasing towards the bottom waters. In contrast, bicarbonate (HCO₃⁻) exhibited more variable vertical behaviour among the

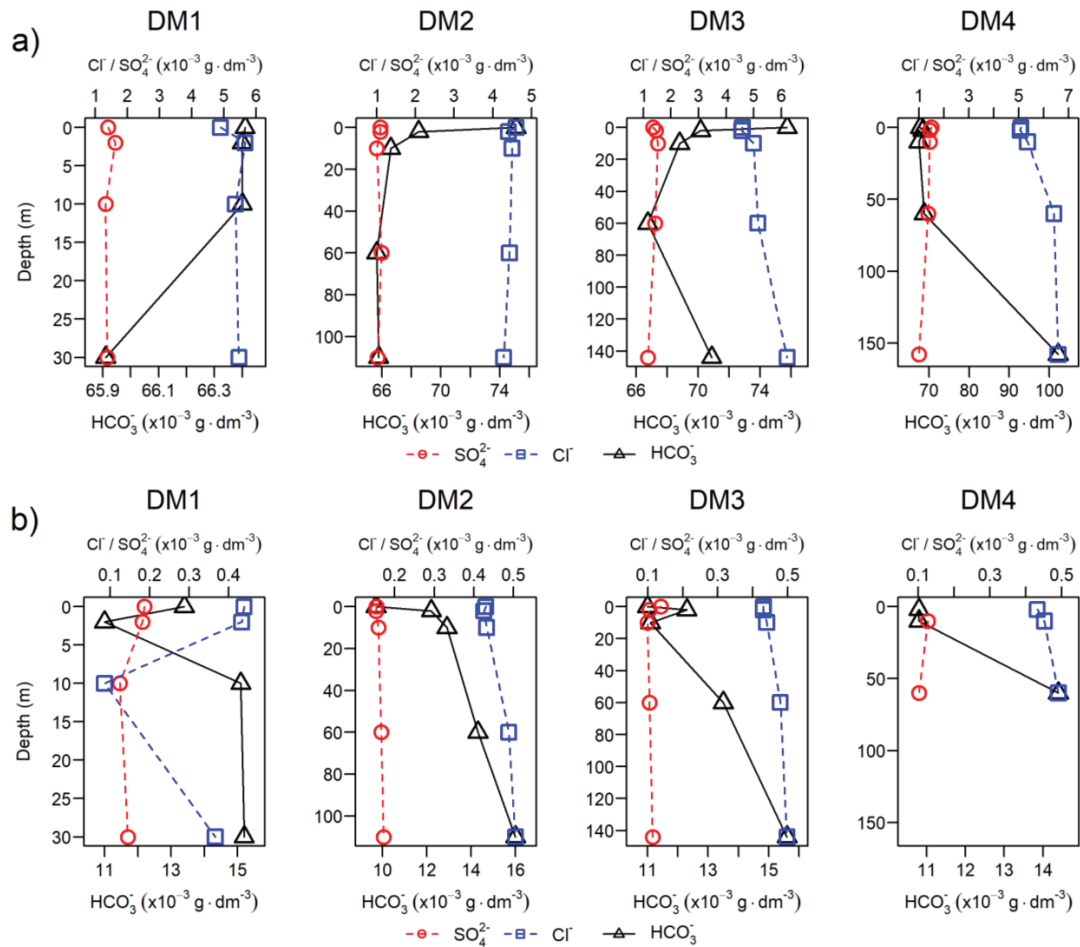


Fig. 3. Anion profiles in water column in: a) 2022, b) 2023; DM1–DM4 = stations; source: own study

surveys. In November 2022, HCO_3^- concentrations decreased with depth at DM1 and DM2 stations but accumulated in deeper layers at DM3 and DM4, indicating localised enrichment under less-uniform vertical mixing. In contrast, in August 2023, the HCO_3^- concentrations generally increased with depth across all stations, consistent with stronger and more persistent stratification.

Statistical analyses relative to the thermocline revealed contrasting spatial patterns among anions and between the surveys. In November 2022, SO_4^{2-} concentrations differed significantly among stations above the thermocline ($F = 10.48$, $p = 0.023$), but not below ($F = 2.496$, $p = 0.134$). Conversely, in August 2023, no significant differences were observed above the thermocline ($F = 2.871$, $p = 0.126$), whereas significant differences were observed below it ($F = 45.42$, $p = 0.021$). A similar depth-dependent pattern was found for Cl^- , with significant inter-station differences only below the thermocline in November 2022 ($F = 6.695$, $p = 0.014$) and August 2023 ($F = 27.36$, $p = 0.035$). Post-hoc tests (Tab. S2) indicated that DM2 station consistently showed lower SO_4^{2-} and Cl^- concentrations than the other stations above and below the thermocline in November 2022. In August 2023, SO_4^{2-} at DM2 differed from that at DM3 and DM4, whereas Cl^- at DM1 was higher than that at the other stations.

In contrast, HCO_3^- concentrations showed no significant spatial differences among stations above or below the thermocline during both surveys (November 2022: above, $F = 2.001$, $p = 0.256$;

below, $F = 1.257$, $p = 0.352$; August 2023: above, $F = 0.899$, $p = 0.494$; below, $F = 1.125$, $p = 0.937$), despite its pronounced vertical variability. Overall, these results indicate that SO_4^{2-} and Cl^- exhibited depth-dependent spatial heterogeneity linked to stratification intensity, whereas HCO_3^- variability was primarily controlled by vertical processes rather than horizontal station differences.

The vertical distribution of sulphide (HS^-) showed pronounced temporal variability between the surveys (Fig. 4). Bottom-layer HS^- concentrations at station DM2 in November 2022 were unavailable because of rapid oxidation during sample preparation. In August 2023, stable stratification promoted HS^- accumulation in the hypolimnion, whereas in November 2022, distinct vertical stratification was observed at all the stations. Overall, HS^- concentrations were higher in August 2023 (from $0.0005 \cdot 10^{-3}$ to $0.4175 \cdot 10^{-3}$ g·dm $^{-3}$) than in November 2022 (from $0.001 \cdot 10^{-3}$ to $0.335 \cdot 10^{-3}$ g·dm $^{-3}$), with maximum values observed at DM1 ($0.410 \cdot 10^{-3}$ g·dm $^{-3}$) and DM4 ($0.335 \cdot 10^{-3}$ g·dm $^{-3}$), respectively.

When evaluated relative to the thermocline, HS^- displayed contrasting spatial patterns between surveys. In November 2022, HS^- concentrations above the thermocline differed significantly among the stations ($F = 458.2$, $p < 0.001$), whereas no significant differences were detected below the thermocline ($F = 1.009$, $p = 0.462$). In contrast, during August 2023, HS^- showed no significant spatial differences either above ($F = 0.901$, $p = 0.494$) or below the thermocline ($F = 12.39$, $p = 0.075$).

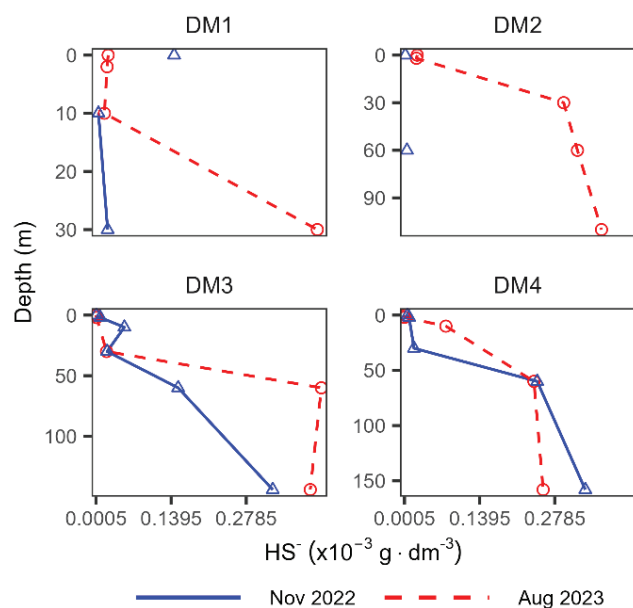


Fig. 4. Vertical profiles of HS^- concentration in the water column in DM1, DM2, DM3 and DM4 stations in November 2022 and August 2023; source: own study

WATER TYPE CLASSIFICATION

Each anion in each measurement was plotted on a ternary diagram (Giggenbach *et al.*, 1983). The ternary diagram illustrates the relative proportions of Cl^- , SO_4^{2-} , and HCO_3^- from the sampling stations during the November 2022 and August 2023 surveys to characterise the hydrochemical composition of Lake Maninjau (Fig. 5). The diagram reveals a consistent dominance of HCO_3^- in both datasets, with a slight increase in the relative proportion of HCO_3^- observed in August 2023. In both surveys, the data points were predominantly clustering close to the 90% HCO_3^- apex. The distribution of data points in the ternary diagram indicates a bicarbonate-dominated hydrochemical composition with only minor contributions from Cl^- and SO_4^{2-} . This pattern further supports the bicarbonate-dominated hydrochemical regime of the lake. These results highlight the stability of the hydrochemical composition between the two survey periods, with bicarbonate consistently dominating the anionic profile.

ELEMENTAL COMPOSITION, SULPHUR AND CARBON CONTENT IN DM1 AND DM3 STATIONS

As shown in Figure 6, silica is the most abundant element in sediments from both stations, with concentrations ranging from 62.7 to 75.8 wt% at DM3 and from 71.5 to 77.8 wt% at DM1. Silica is consistently distributed throughout all the sediment layers at both sites. Other major elements included Al, Fe, P, S, Mg, Ca, K, and Mn. Some elements (Al, Fe, P, Mg, Ca, and K) were more enriched in DM1 sediments, with concentrations of 11.10 wt%, 8.96 wt%, 1.96 wt%, 1.80 wt%, 1.26 wt%, and 0.60 wt%, respectively, compared to those in DM3, which exhibited lower concentrations of 10.08 wt%, 6.73 wt%, 0.81 wt%, 1.61 wt%, 0.75 wt%, and 0.5 wt%. In contrast, sulphur (S) was more abundant in DM3 sediments than in DM1, with concentrations of 4.60 wt% and 1.86 wt%, respectively.

The sediment cores from DM3 and DM1 stations can be clearly separated based on their distribution in the PCA biplots (Fig. 7). Samples from DM3 predominantly clustered on the left-hand side of the biplot, whereas DM1 samples were mainly grouped on the right side. The PCA results indicate that the DM1 sediments are dominated by lithogenic inputs, as reflected by their strong association with Al, K, and Ti. In contrast, sediments from station DM3 were characterised by higher sulphur content and a relatively quartz-siliceous composition (Si-rich). Notably, organic carbon (Corg) exhibited a loading direction similar to that of sulphur along both PC1 and PC2, indicating a close positive association between these variables in the PCA space. According to Wils *et al.* (2021), the enrichment of silica at DM3 may reflect a strong biogenic contribution.

The total carbon (C) and sulphur (S) contents in sediments at DM1 and DM3 stations are shown in Table S3. The total carbon (TC) concentrations in DM3 sediments ranged from 8.947%wt to 14.079%wt, while the total sulphur (TS) ranged between 0.877%wt and 1.464%wt. In DM1 sediments, TC increased towards the sediment surface, likely reflecting enhanced organic matter inputs associated with floating net cage aquaculture, whereas TS concentrations varied with depth.

The relationship between the total carbon and total sulphur content in the sediments from Lake Maninjau is illustrated in Figure 8. Sediment samples from DM1 station (nearshore eutrophic

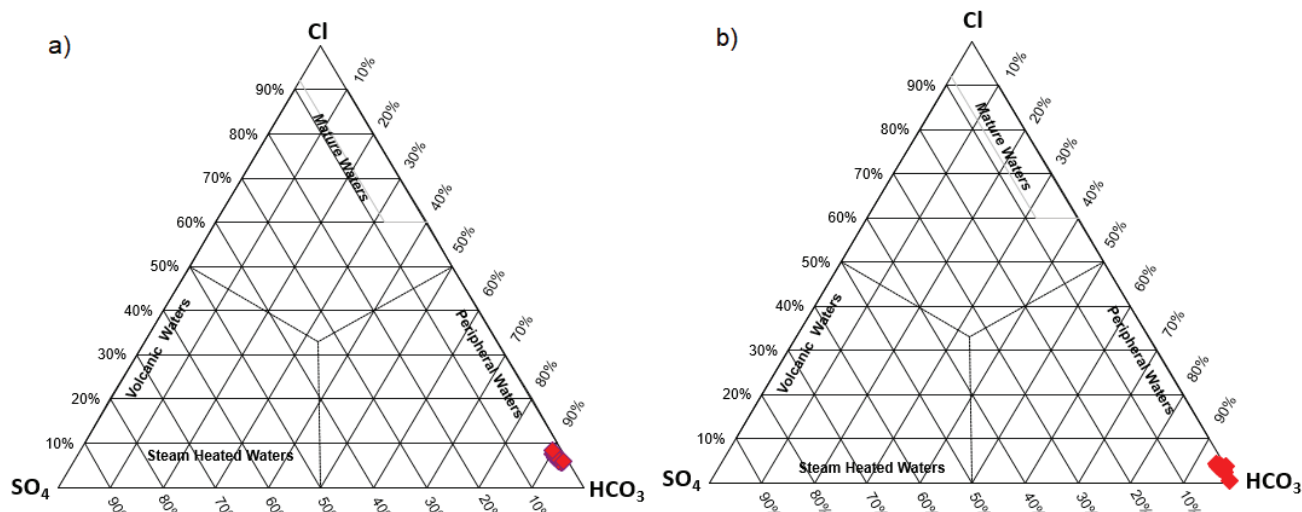


Fig. 5. Ternary diagram $\text{Cl-SO}_4\text{-HCO}_3$ in: a) November 2022, b) August 2023; source: own study

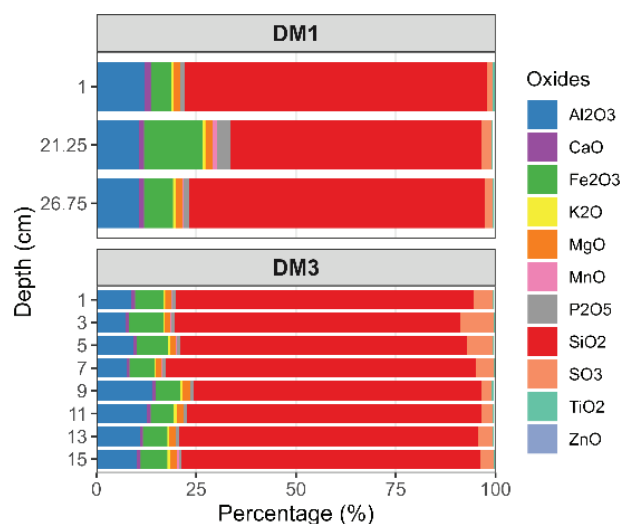


Fig. 6. Elemental composition in DM3 and DM1 sediments of Lake Maninjau; source: own study

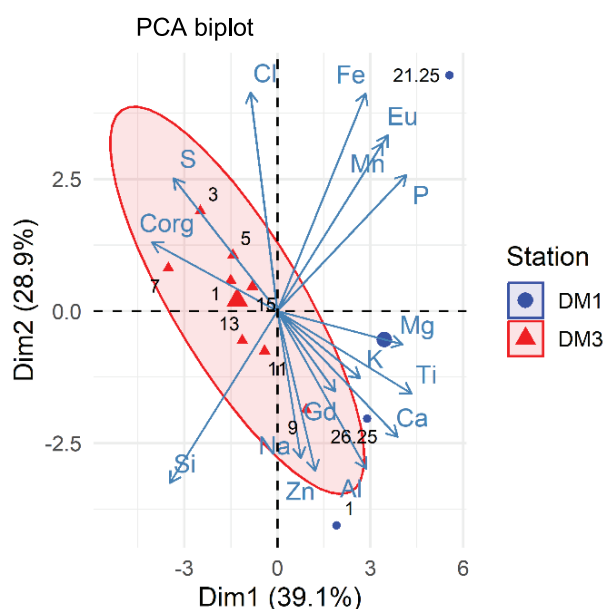


Fig. 7. Principal component analysis (PCA) of elemental composition in DM3 and DM1 sediments of Lake Maninjau; source: own study

zone) and DM3 (dome-proximal deep area) exhibited a statistically significant positive correlation between C and S (Pearson's $r = 0.73$, $p = 0.01$), following a linear trend with an R^2 value of 0.54. This pattern indicates a moderate to strong coupling between organic carbon accumulation and sulphur enrichment in sediments. Compared to DM1, the DM3 samples displayed a wider range of TC and TS values, whereas the DM1 samples clustered toward the lower end of the observed concentration range.

DISCUSSION

TRANSITION FROM VOLCANIC TO BIOTIC CONTROL

Although Lake Maninjau originated from a caldera-forming eruption, the hydrogeochemical and thermal data from all stations (DM1, DM2, DM3, and DM4) provide no evidence of active volcanic or hydrothermal influence. During the November

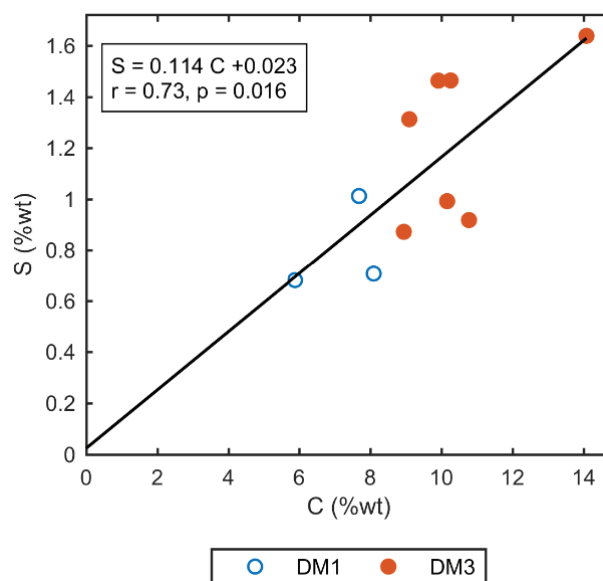


Fig. 8. Relationship between total sulphur (S) and total carbon (C) in DM3 and DM1 sediments of Lake Maninjau; source: own study

2022 survey, temperature differences above and below the thermocline varied among stations but remained very small, with surface and bottom-water differences ranging from 0.11–0.86°C and 0.008–0.038°C, respectively. These minor temperature gradients do not represent thermal anomalies and are more likely related to differences in sampling time rather than subsurface heat input. Furthermore, uniform bottom-water temperatures across the basin, including the dome-proximal area (DM3), further support this interpretation. This contrasts with active volcanic lakes such as Copahue or Ranau, where strong thermal gradients reflect geothermal activity (Zaennudin *et al.*, 2011; Tamburello *et al.*, 2015; Candela-Becerra *et al.*, 2020). Therefore, the slightly higher water temperatures observed in August 2023 are more plausibly attributed to solar heating during dry conditions rather than volcanic forcing (Putri *et al.*, 2024).

Across all sampling stations, the pH of the water column remained neutral to alkaline, which is consistent with carbonate-buffered systems rather than acid-sulphate volcanic lakes. The slightly higher pH values at DM1 compared to the deeper dome area (DM3) are likely attributable to intense photosynthetic uptake of CO_2 by phytoplankton, driven by nutrient enrichment from aquaculture activities (Cynthia Henny and Nomosatryo, 2016). The total dissolved solids (TDS) profile showed low and consistent values across all stations, with a minor increase in the deeper layers. Such low TDS values ($<10 \text{ g}\cdot\text{dm}^{-3}$) are characteristic of a low-activity volcanic lake (Rouwet *et al.*, 2014), demonstrating that volcanic activity does not primarily control sulphide dynamics. Instead, the observed patterns are more consistent with natural stratification processes rather than geothermal enrichment, aligning with findings from other post-volcanic tropical lakes, such as Singkarak (Fukushima *et al.*, 2021).

The anion composition across sites was consistently dominated by bicarbonate, confirming a CO_2 -buffered, low-salinity system. The lack of chloride and sulphate enrichment and the absence of acidic conditions distinguish Lake Maninjau from active crater lakes such as Kawah Ijen and Quilotoa (Aguilera *et al.*, 2000). Collectively, the spatially uniform temperature, stable ionic composition, and bicarbonate dominance demon-

strate that Lake Maninjau has transitioned from a volcanically influenced lake to a system controlled by biological and seasonal processes.

BIOTIC AND ANTHROPOGENIC DRIVERS OF SULPHIDE FORMATION

Comparisons between sediment cores from DM3 (near the volcanic dome) and DM1 further support the predominance of biotic processes in sulphide production. Although DM3 lies closest to the dome and might therefore be expected to exhibit volcanic input, its sediment characteristics is more consistent with active microbial sulphate reduction (MSR) rather than hydrothermal deposition. The strong positive correlation between total sulphur and carbon and the positive C-S intercept plots (Fig. 8), along with the relative depletion of porewater sulphate (Fig. S2), reflect the bacterial conversion of sulphate to sulphide and its subsequent incorporation into the solid phase (Ding *et al.*, 2016).

Although DM3 and DM4 stations are located far from direct organic input sources, hydrogen sulphide (HS^-) concentrations in the hypolimnion remain high, indicating that sulphide production is not limited to nearshore zones. In these deeper central stations, organic material can still accumulate owing to geochemical focusing processes that transport and concentrate particulate matter toward the deeper basin. Previous studies have shown that organic matter content in deeper lake sediments is often higher than that in shallower areas as a result of focusing and redeposition mechanisms (Henny, 2009; Dianto, Ridwansyah and Subehi, 2020). Persistently anoxic conditions in the hypolimnion further support microbial sulphate reduction, thereby enhancing sulphide production. At the same time, a portion of the sulphide formed in the deep basin may be removed from the water column through reactions with reactive iron, leading to the formation of iron sulphides. This mechanism is supported by XRF results showing higher Fe and S contents in surface sediments at DM3 (4.9% Fe and 2.0% S) compared to DM1 (3.5% Fe and 0.5% S), which helps explain why sulphide concentrations at DM3 ($0.397 \cdot 10^{-3} \text{ g} \cdot \text{dm}^{-3}$) are comparable to those at DM1 ($0.413 \cdot 10^{-3} \text{ g} \cdot \text{dm}^{-3}$). Overall, these patterns indicate that sulphide dynamics in Lake Maninjau are governed by basin-scale particle focusing, persistent hypolimnetic anoxia, and iron-sulphur interactions, which regulate sulphide retention in sediments. Similar linkages between enhanced organic accumulation, microbial sulphate reduction, and sulphide enrichment have also been reported in other tropical and subtropical lakes, such as Lake Bhimtal (Choudhary, Routh and Chakrapani, 2013), Nakaumi (Sakai *et al.*, 2013), and even in previous studies of Maninjau itself (Henny, 2009; Henny and Nomosatryo, 2012).

These patterns demonstrate that the sulphur cycle of Lake Maninjau is driven primarily by organic matter degradation and microbial metabolism, with little evidence for a significant contribution from volcanic or magmatic gases. The lake's morphology remains volcanic, but its geochemical function is now shaped by anthropogenic eutrophication and biogeochemical feedback.

INTEGRATED HYDROGEOCHEMICAL INTERPRETATION

Synthesising the hydrogeochemical evidence, Lake Maninjau is characterised as a carbonate-buffered, biologically regulated, post-volcanic lake. The dominance of HCO_3^- indicates a carbonate-

buffered system primarily controlled by the dissolution of atmospheric and biogenically derived CO_2 , which is consistent with the $\text{pH}-(\text{Cl}^- + \text{SO}_4^{2-})$ relationship (Fig. S3). Both lake water and sediment porewater plot within the CO_2 -dominated field (pH 6–9; combined anion $<10 \text{ mg} \cdot \text{dm}^{-3}$), confirming equilibrium with carbonate buffering rather than the influence of acidic magmatic gases such as H_2S , SO_2 , HCl , or HF (Varekamp, Pasternack and Rowe Jr., 2000). This interpretation is further supported by the XRF analysis results, which show no detectable enrichment of trace elements commonly used as indicators of volcanic or hydrothermal inputs, such as boron (B), zinc (Zn), arsenic (As), molybdenum (Mo), tin (Sn), antimony (Sb), thallium (Tl), and lead (Pb), as proposed by Keizer (2014) – Table S4. The absence of magmatic signals, stable ionic composition, and evidence of microbial sulphate reduction collectively indicate that the lake's geochemical behaviour is governed by internal biogeochemical processes.

Seasonal variations further supports this interpretation: during the dry season (August 2023), stronger thermal stratification promoted sulphide accumulation in the hypolimnion, whereas in the wet season (November 2022), vertical mixing resulted in more homogenised sulphide concentrations at shallower sites such as DM1. Elevated HCO_3^- concentrations in the wet season likely reflect intensified microbial respiration and organic matter degradation, which increase dissolved CO_2 concentrations in the water column. Collectively, the coupling of carbonate buffering, biological metabolism, and seasonal stratification-mixing dynamics demonstrates a self-sustaining biogeochemical cycle typical of eutrophic tropical lakes, where sulphide generation is primarily biologically mediated rather than driven by residual geothermal inputs (Duarte and Prairie, 2005; Katsev *et al.*, 2010).

IMPLICATIONS OF THE FINDINGS

The findings of this study provide important scientific insights into the geochemical evolution of volcanic-caldera lakes. The results demonstrate that volcanic lakes, such as Lake Maninjau, can undergo a complete transition from magmatic to biologically dominated systems, offering a conceptual model for understanding post-volcanic lake succession (Rouwet *et al.*, 2014). Through the integration of hydrogeochemical and sedimentary analyses, this study established a reliable framework for distinguishing between volcanic and biotic sources of sulphur, which may be applicable to other CO_2 -dominated lake systems, including Lakes Singkarak and Matano (Crowe *et al.*, 2011; Fukushima *et al.*, 2021). This study also underscores the pivotal role of microbial processes in regulating sulphur cycling under eutrophic conditions. This highlights how biological activity, particularly microbial sulphate reduction, can significantly alter sulphur dynamics and redox conditions, thereby enriching our understanding of biogeochemical transformations in tropical aquatic systems.

From an environmental management perspective, these findings emphasise the need to mitigate anthropogenic pressures that promote sulphide accumulation. Management strategies should prioritise reducing organic matter and nutrient inputs from aquaculture activities, as excessive nutrient inputs enhance microbial sulphate reduction and exacerbates hypoxic conditions. Establishing a long-term geochemical monitoring program that

tracks redox-sensitive species, such as sulphide, iron, and manganese, would allow for the early detection of ecosystem degradation and timely intervention. Furthermore, the hydrogeochemical approach presented in this study serves as a valuable reference model for the restoration and risk assessment of other post-volcanic lakes facing similar anthropogenic influence.

CONCLUSIONS

The hydrogeochemical and sedimentary data indicate that sulphide dynamics in Lake Maninjau are predominantly controlled by biochemical and anthropogenic processes with little evidence for a significant contemporary contribution from volcanic or hydrothermal activity. The neutral to alkaline pH, low total dissolved solids, and bicarbonate-dominated water chemistry indicate a carbonate-buffered system that is consistent with biologically regulated processes. The positive relationship between sedimentary sulphur and carbon further supports sulphide production, primarily through microbial sulphate reduction coupled with organic matter degradation.

SUPPLEMENTARY MATERIALS

Supplementary material to this article can be found online at: https://www.jwld.pl/files/Supplementary_material_70_Jasalesmana.pdf.

FUNDING

This work was supported by the Saintek Scholarship Program, Ministry of Research and Technology/National Research and Innovation Agency (BRIN) and Research Program of the Ministry of Education, Culture, Research, and Technology 2023, contract number: 110/E5/PG.02.00.PL/2023.

CONFLICT OF INTERESTS

All authors declare that they have no conflict of interests.

REFERENCES

- Aguilera, E. *et al.* (2000) "Water chemistry of Lake Quilotoa (Ecuador) and assessment of natural hazards," *Journal of Volcanology and Geothermal Research*, 97(1–4), pp. 271–285. Available at: [https://doi.org/10.1016/S0377-0273\(99\)00168-7](https://doi.org/10.1016/S0377-0273(99)00168-7).
- Alloway, B.V. *et al.* (2004) "Correspondence between glass-FT and ¹⁴C ages of silicic pyroclastic flow deposits sourced from Maninjau caldera, west-central Sumatra," *Earth and Planetary Science Letters*, 227(1–2), pp. 121–133. Available at: <https://doi.org/10.1016/j.epsl.2004.08.014>.
- Candela-Becerra, L.J. *et al.* (2020) "Thermal evolution of the Crater Lake of Copahue Volcano with ASTER during the last quiescence period between 2000 and 2012 eruptions," *Journal of Volcanology and Geothermal Research*, 392, 106752. Available at: <https://doi.org/10.1016/j.jvolgeores.2019.106752>.
- Choudhary, P., Routh, J. and Chakrapani, G.J. (2013) "A 100-year record of changes in organic matter characteristics and productivity in Lake Bhimtal in the Kumaon Himalaya, NW India," *Journal of Paleolimnology*, 49(2), pp. 129–143. Available at: <https://doi.org/10.1007/s10933-012-9647-9>.
- Crowe, S.A. *et al.* (2011) "The methane cycle in ferruginous Lake Matano," *Geobiology*, 9(1), pp. 61–78. Available at: <https://doi.org/10.1111/j.1472-4669.2010.00257.x>.
- Dianto, A., Ridwansyah, I. and Subehi, L. (2020) "Organic matter and organic carbon levels in sediments of Lake Maninjau, West Sumatra," *IOP Conference Series: Earth and Environmental Science*, 535, 012030. Available at: <https://doi.org/10.1088/1755-1315/535/1/012030>.
- Ding, X. *et al.* (2016) "Sulfur geochemistry of a lacustrine record from Taiwan reveals enhanced marine aerosol input during the Early Holocene," *Scientific Reports*, 6, 38989. Available at: <https://doi.org/10.1038/srep38989>.
- DKPP (2024) *Data produksi budidaya di KJA Kabupaten Agam tahun 2020–2024 [Data on cultivation production in Floating Net Cages Agam Regency in 2020–2024]*. Lubuk Basung: Kabupaten Agam.
- Duarte, C.M. and Prairie, Y.T. (2005) "Prevalence of heterotrophy and atmospheric CO₂ emissions from aquatic ecosystems," *Ecosystems*, 8(7), pp. 862–870. Available at: <https://doi.org/10.1007/s10021-005-0177-4>.
- Fukushima, T. *et al.* (2017) "Will hypolimnetic waters become anoxic in all deep tropical lakes?," *Scientific Reports*, 7, 45320. Available at: <https://doi.org/10.1038/srep45320>.
- Fukushima, T. *et al.* (2021) "Convection of waters in Lakes Maninjau and Singkarak, tropical oligomictic lakes," *Limnology*, 23, pp. 375–383. Available at: <https://doi.org/10.1007/s10201-021-00686-8>.
- Giggenbach, W.F. *et al.* (1983) "Isotopic and chemical composition of Parbati Valley geothermal discharges, North-West Himalaya, India," *Geothermics*, 12(2), pp. 199–222. Available at: [https://doi.org/10.1016/0375-6505\(83\)90030-5](https://doi.org/10.1016/0375-6505(83)90030-5).
- Henny, C. (2009) "Dynamics of biogeochemistry of sulfur in Lake Maninjau," *Limnotek perairan darat tropis di Indonesia*, 16(2), pp. 74–87.
- Henny, C. and Nomosatryo, S. (2012) "Sulfide dynamics in Lake Maninjau: The implication on phosphate release in the hypolimnion," *Limnotek: perairan darat tropis di Indonesia*, 19(2), pp. 102–112.
- Henny, C. and Nomosatryo, S. (2016) "Changes in water quality and trophic status associated with cage aquaculture in Lake Maninjau, Indonesia," *IOP Conference Series: Earth and Environmental Science*, 31(1), 012027. Available at: <https://doi.org/10.1088/1755-1315/31/1/012027>.
- Jasalesmana, T., Putri, M.R. and Abdurrachman, M. (2023) "Water quality assessment of Lake Maninjau after the mass fish kill event," *IOP Conference Series: Earth and Environmental Science*, 1245, 012013. Available at: <https://doi.org/10.1088/1755-1315/1245/1/012013>.
- Junaidi *et al.* (2022) "Floating cage aquaculture production in Indonesia: Assessment of opportunities and challenges in Lake Maninjau," *AIMS Environmental Science*, 9(1), pp. 1–15. Available at: <https://doi.org/10.3934/environsci.2022001>.
- Kankaala, P. *et al.* (2013) "Variable production by different pelagic energy mobilizers in boreal lakes," *Ecosystems*, 16, pp. 1152–1162. Available at: <https://doi.org/10.1007/s10021-013-9674-z>.
- Katsev, S. *et al.* (2010) "Mixing and its effects on biogeochemistry in the persistently stratified, deep, tropical," *Limnology and Oceanography*

- graphy, 55(2), pp. 763–776. Available at: <https://doi.org/https://doi.org/10.4319/lo.2010.55.2.0763>.
- Keizer, M. (2014) *Origin and significance of mobile trace elements in volcanic lakes – A case study at Poás, Costa Rica*. MSc Thesis. Universiteit Utrecht. Available at: <https://studenttheses.uu.nl/server/api/core/bitstreams/cba82a74-8cb1-4673-801d-a46e346beb1c/content> (Accessed: October 25, 2025).
- Kusakabe, M. (2015) “Evolution of CO₂ content in Lakes Nyos and Monoun, and sub-lacustrine CO₂-recharge system at Lake Nyos as envisaged from CO₂/³He ratios and noble gas signatures,” in D. Rouwet *et al.* (eds.) *Volcanic lakes*. Berlin–Heidelberg: Springer-Verlag, pp. 427–450. Available at: https://doi.org/10.1007/978-3-642-36833-2_19.
- Makmur, S., Muthmainnah, D. and Subagdja (2020) “Fishery activities and environmental condition of Maninjau Lake, West Sumatra,” *IOP Conference Series: Earth and Environmental Science*, 564(1), 012025. Available at: <https://doi.org/10.1088/1755-1315/564/1/012025>.
- Nomosatryo, S. *et al.* (2021) “Geochemical characteristics of sediment in tropical Lake Sentani, Indonesia, are influenced by spatial differences in catchment geology and water column stratification,” *Frontiers in Earth Science*, 9, 671642. Available at: <https://doi.org/10.3389/feart.2021.671642>.
- Pearson, F. (1981) “Fixed endpoint alkalinity determination,” *Water Pollution Control Federation*, 53(7), pp. 1243–1252. Available at: <http://www.jstor.org/stable/25041459>. (Accessed: October 25, 2025).
- Powell, T. and Cumming, W. (2010) “Spreadsheet for geothermal water and gas geochemistry,” in *Proceedings, Thirty-Fifth Workshop on Geothermal Reservoir Engineering*, SGP-TR-188, California, Stanford, CA, 1–3 Feb 2010. Stanford: Stanford University, pp. 1–10. <https://earthsciences.stanford.edu/ERE/pdf/IGASTandard/SGW/2010/powell.pdf> (Accessed: October 25, 2025).
- Pribadi, A., Mulyadi, E. and Pratomo, I. (2007) “Mekanisme erupsi ignimbrit kaldera Maninjau, Sumatera Barat [The mechanism of ignimbrite eruption in Maninjau caldera, West Sumatra],” *Jurnal Geologi Indonesia*, 2(1), pp. 31–41. Available at: <https://doi.org/10.17014/ijog.2.1.31-41>.
- Putri, M.R. *et al.* (2024) “The impact of weather condition changes on vertical distribution of sulfides in Lake Maninjau based on observation data,” *Limnotek: perairan darat tropis di Indonesia*, 1(2). Available at: <https://doi.org/10.55981/limnotek.2024.2203>.
- Rouwet, D. *et al.* (2014) “Past, present and future of volcanic lake monitoring,” *Journal of Volcanology and Geothermal Research*, 272, pp. 78–97. Available at: <https://doi.org/10.1016/j.jvolgeores.2013.12.009>.
- Rustini, H.A., Harsono, E. and Ridwansyah, I. (2019) “Modeling the influence of floating net aquaculture and nutrient loads from catchment on the trophic status of a tropical volcanic lake,” in *IOP Conference Series: Earth and Environmental Science*, 535, 012031. Available at: <https://doi.org/10.1088/1755-1315/535/1/012031>.
- Sakai, S. *et al.* (2013) “Hydrogen sulfide and organic carbon at the sediment – water interface in coastal brackish Lake Nakaumi, SW Japan,” *Environmental Earth Sciences*, 68, pp. 1999–2006. Available at: <https://doi.org/10.1007/s12665-012-1887-5>.
- Santoso, A.B. *et al.* (2018) “What do we know about Indonesian tropical lakes? Insights from high frequency measurement,” *IOP Conference Series: Earth and Environmental Science*, 118(1), 012024. Available at: <https://doi.org/10.1088/1755-1315/118/1/012024>.
- Santoso, A.B. and Triwisesa, E. (2020) “Ecosystem metabolism and oxygen deficit in Lake Maninjau: Insight from high-frequency measurement,” *Limnotek: perairan darat tropis di Indonesia*, 27(2), pp. 93–102. Available at: <https://doi.org/10.14203/limnotek.v27i2.306>.
- Setiawan, F. *et al.* (2019) “Long-term change of the secchi disk depth in Lake Maninjau, Indonesia shown by landsat TM and ETM+ data,” *Remote Sensing*, 11(23), 2875. Available at: <https://doi.org/10.3390/rs11232875>.
- Sulastri, Aisyah, S. and Dina, R. (2022) “Trophic status and phytoplankton community structure of four small lakes in Ciliwung Watershed, Indonesia,” in *IOP Conference Series: Earth and Environmental Science*, 950, 012037. Available at: <https://doi.org/10.1088/1755-1315/950/1/012037>.
- Syandri, H. *et al.* (2014) “State of aquatic resources Maninjau Lake West Sumatra Province, Indonesia,” *International Journal of Ecology and Environmental Sciences*, 5(1), pp. 109–113. Available at: https://bioinfopublication.org/files/articles/5_1_1_JEES.pdf (Accessed: October 25, 2025).
- Tamburello, G. *et al.* (2015) “Intense magmatic degassing through the lake of Copahue volcano, 2013–2014,” *Journal of Geophysical Research, Solid Earth*, 120, pp. 6071–6084. Available at: <https://doi.org/10.1002/2015JB012160>.
- Varekamp, J.C., Pasternack, G.B. and Rowe Jr., G.L. (2000) “Volcanic lake systematics II. Chemical constraints,” *Journal of Volcanology and Geothermal Research*, 97, pp. 161–179. Available at: [https://doi.org/10.1016/S0377-0273\(99\)00182-1](https://doi.org/10.1016/S0377-0273(99)00182-1).
- Wils, K. *et al.* (2021) “The sediments of Lake Singkarak and Lake Maninjau in West Sumatra reveal their earthquake, volcanic and rainfall history,” *Sedimentary Geology*, 416, 105863. Available at: <https://doi.org/10.1016/j.sedgeo.2021.105863>.
- Zaennudin, A. *et al.* (2011) “Studi awal fenomena kematian ikan di Danau Ranau, Sumatra Selatan [Initial study of the fish mortality phenomenon in Lake Ranau, South Sumatra],” *Jurnal Lingkungan dan Bencana Geologi*, 2(2), pp. 77–94. <https://doi.org/10.34126/jlbg.v2i2.23>.